Anthony Lieu

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Education

Rochester Institute of Technology — Rochester, NY
Bachelor of Science in Software Engineering — *Expected Summer 2027*
- GPA: 3.89/4.0

Experience

Software Engineer Co-op — May 2026 – Present
Paychex — *Rochester, NY*
- Automated the migration of 400+ repositories across multiple departments to new OpenShift CRD versions, eliminating 20+ hours of manual manifest updates.
- Resolved an infinite heap-dump loop in the OpenShift base image health monitor and hardened restart logic against transient errors, improving stability for apps built on the shared Java base images.
- Introduced v2 OpenShift application and project CRDs with build-time policy checks across the onboarding service and Gradle plugin, preventing misconfigured tags from reaching deployment.
- Fixed the XL Release secret rotation template rotating only the first account and added parallel execution for up to 10 accounts, ensuring every account's secrets rotate in a single run.
- Reduced the team support queue backlog by resolving 6–8 issues per weekly rotation, including build and deployment failures, cutting developer wait time for answers by 50%.

Software Engineer Co-op — Jan. 2026 – May 2026
RIT Department of Software Engineering — *Rochester, NY*
- Built 15+ CRUD API endpoints for projects, project proposals, interest forms, and the applicant dashboard, enabling coordinators and supervisors to manage the co-op project lifecycle in one place.
- Modeled the database schema for the co-op program's entities, including projects, proposals, interest forms, and applicants, translating product requirements into a normalized structure of related tables.
- Expanded role-based access control across 6 user roles, restricting API access and conditionally rendering UI features based on each role's permissions so users only see and do what their role allows.
- Developed the frontend dashboard for these features, including the applicant view, with table, card, and kanban views so users can work in the layout that suits them.

Software Engineer in Test Co-op — Jun. 2025 – Aug. 2025
Excellus BlueCross BlueShield — *Rochester, NY*
- Achieved 90–100% test coverage across 6 shared libraries, up from 30–40%, by building unit and integration test suites that covered previously untested code paths relied on in production by downstream teams.
- Diagnosed a production logging defect by tracing masked error types through shared library code, then patched the root cause to restore accurate error reporting across all dependent engineering systems.
- Added 120+ test cases across UI and API automation codebases and ran regression testing after changes to confirm UI workflows and API behavior remained stable.
- Authored changelogs and revised README, internal, and SharePoint documentation, keeping references accurate for the engineers consuming the shared libraries.

Projects

Intelligent Stock Market News Analyzer | *Python, React, MySQL, AWS* — Sep. 2025 – Dec. 2025
- Architected a serverless backend that decouples ingestion, sentiment processing, and data serving, keeping the dashboard responsive during heavy ingestion, and automated environment setup and teardown to remove manual deployment steps.
- Implemented real-time news ingestion and sentiment scoring for the Nasdaq-100, loading about 1,000 articles initially then processing roughly 10 per stock per minute for frontend consumption.
- Designed interactive sentiment charts across multiple timeframes, including hourly, daily, and weekly, with a companion price chart over the same period so users can compare sentiment shifts against price movement.
- Created an email notification feature that alerts users whenever sentiment changes for a stock on their watchlist, so they can react without constantly monitoring the market.

Technical Skills

Languages: Java, Python, C, TypeScript/JavaScript, Groovy, Kotlin, Rego, HTML/CSS
Frameworks & Libraries: React, Node.js, Next.js, Express.js, Angular, Spring Boot, Flask, JUnit, Pytest, Selenium
Databases & ORMs: PostgreSQL, MySQL, MariaDB, Prisma
Cloud & DevOps: AWS, Docker, Kubernetes, OpenShift, Terraform, GitHub Actions, Jenkins, Gradle, XL Release, XL Deploy, Artifactory, Open Policy Agent